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13.1.1: Fibonacci Sequence using Modules

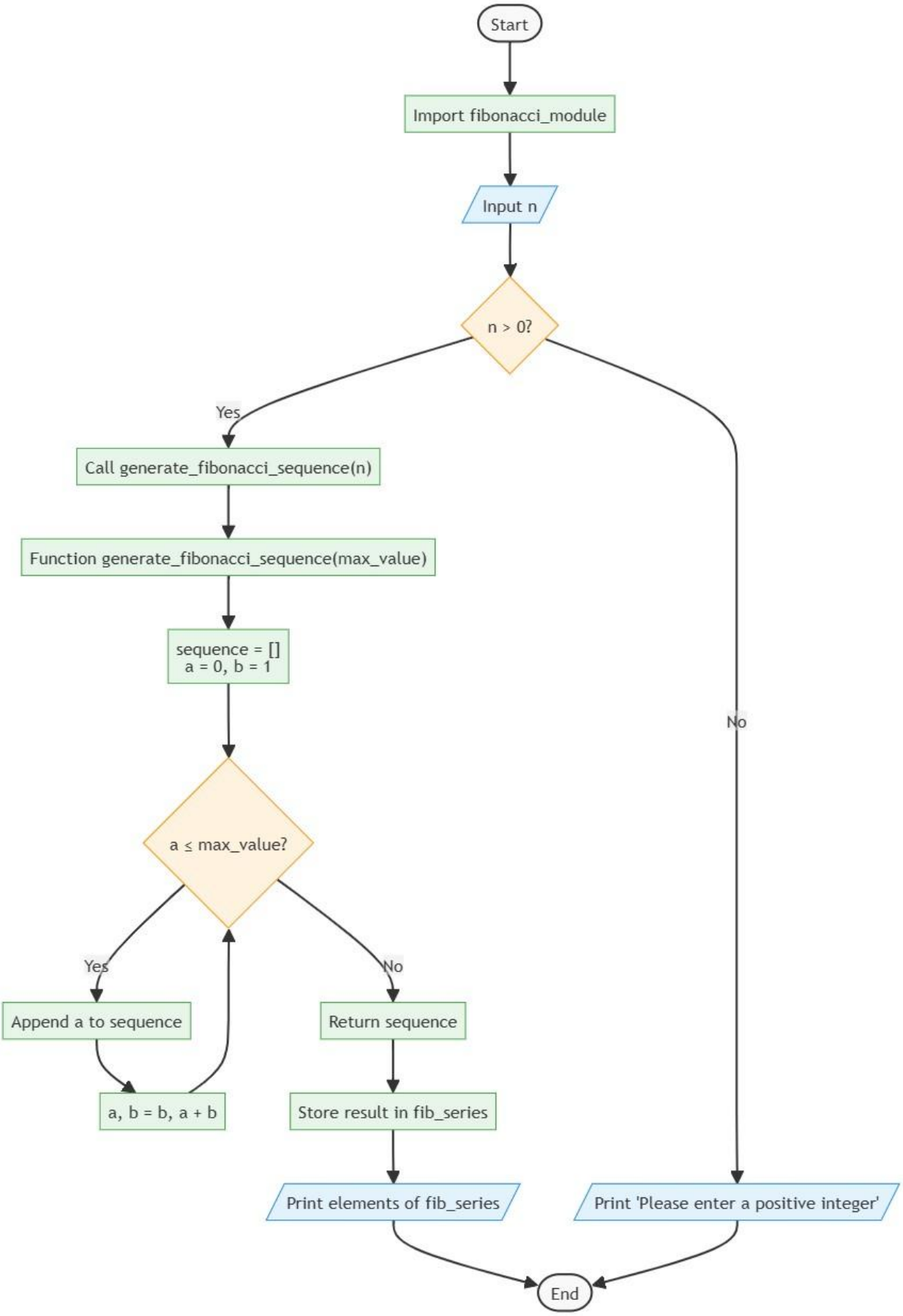
Algorithm:
fibonacci_module.py
Steps:

1. Function definition
   def generate_fibonacci_sequence(max_value):
2. Initialize
   - sequence = [] (empty list to store Fibonacci numbers)
     a, b = 0, 1 (first two Fibonacci numbers)
3. If max_value < 1
   - Return empty list (no valid Fibonacci numbers)
4. Loop while a ≤ max_value:
   - Append a to sequence
   - Update: a, b = b, a + b
5. Return sequence

Algorithm:
main.py
Steps:

1. Import fibonacci_module
2. Read input
   - n = int(input())
3. If n > 0:
   - Call fibonacci_module.generate_fibonacci_sequence(n) → store result in fib_list
   - Print the list as space-separated numbers:
     print(' '.join(map(str, fib_list)))
4. Else (n ≤ 0):
   - Print "Please enter a positive integer"

FLOWCHART:


Start
Import fibonacci_module
Input n
n > 0?
Yes
No
Call generate_fibonacci_sequence(n)
Function generate_fibonacci_sequence(max_value)
sequence = []
a = 0, b = 1
a ≤ max_value?
Yes
No
Append a to sequence
a, b = b, a + b
Return sequence
Store result in fib_series
Print elements of fib_series
Print 'Please enter a positive integer'
End

Code:

Main.py:

```
import fibonacci_module

n = int(input())

if n > 0:
fib_series = fibonacci_module.generate_fibonacci_sequence(n)
print(' '.join(map(str, fib_series)))
else:
print("Please enter a positive integer")
```

fibonacci_module.py

```
def generate_fibonacci_sequence(max_value):
sequence = []

a, b = 0, 1
while a <= max_value:
        sequence.append(a)
        a, b = b, a + b

return sequence
```